Full-corpus results: what we tested and what we found

Run: `2026-11-06_01-23-40-040fd7b7`. Analysis: `full-corpus-report-v1`.

Interactive tables · Case explorer

Main findings

Six models answered the same 54,000 distinct questions. All 108 tasks completed. The corpus has 324,000 saved responses and zero final sample execution errors. There were 288,466 source-label matches, 352 content filters, and 6 other non-interpretable responses across all models. These are separate outcomes. Claude Sonnet 5.5 had the highest observed agreement with source labels: 53,973/53,995 (99.96%). This comparison concerns this corpus and this run. Source labels have not been independently audited for logical validity.

Sonnet 5.5 and Gemini 3.1 Flash-Lite are close in source-label agreement, but differ in format compliance and reported cost. See the tables below for exact counts and missing cost evidence.

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What question does the experiment ask?

We test how six language models answer hypothetical syllogism and metainferential hypothetical syllogism questions under different lexical and grammatical conditions. We compare each answer with the label supplied by the dataset. We also test whether the model follows a strict response-format instruction.

Every case uses a new conversation, one system message, and one original question. No examples are supplied in the prompt (zero-shot). The system message is:

> `Answer only with 'yes' or 'no' and nothing else.`

A case is one complete reasoning question. It can contain several clauses or sentences. We count cases, not individual clauses. The 54,000 question texts are distinct by exact text comparison.

What cases are in the corpus?

Code	Meaning	Cases
HS	Hypothetical syllogism expressed with conditionals	18,000
mHS	Metainferential hypothetical syllogism: relations between inferences	36,000
HF / LF / NS	High-frequency words / low-frequency words / pseudowords	18,000 each
PR / PA / FU	Present / past / future	18,000 each
canonic / non-canonic	Standard / reversed premise order, as labeled by the source	27,000 each
valid / invalid	Source validity label	27,000 each

NS is a pseudoword condition, not another level of word frequency. The filename identifies HS versus mHS; the original internal case-type field is HS in both families. We preserve that field and report family separately.

Each HS dataset has four strata: source validity crossed with premise order. Each mHS dataset has eight strata: those four combinations crossed with two formulations. The formulations are `assuming that we can infer / thus` and `from the inference (therefore) / does the inference follow`. Each stratum contains 500 cases. The complete design has 108 strata.

Dataset	Family	Lexical condition	Tense	Cases	Strata
HS_HF_FU_2000	HS	HF	FU	2000	4
HS_HF_PA_2000	HS	HF	PA	2000	4
HS_HF_PR_2000	HS	HF	PR	2000	4
HS_LF_FU_2000	HS	LF	FU	2000	4
HS_LF_PA_2000	HS	LF	PA	2000	4
HS_LF_PR_2000	HS	LF	PR	2000	4
HS_NS_FU_2000	HS	NS	FU	2000	4
HS_NS_PA_2000	HS	NS	PA	2000	4
HS_NS_PR_2000	HS	NS	PR	2000	4
mHS_HF_FU_4000	mHS	HF	FU	4000	8
mHS_HF_PA_4000	mHS	HF	PA	4000	8
mHS_HF_PR_4000	mHS	HF	PR	4000	8
mHS_LF_FU_4000	mHS	LF	FU	4000	8
mHS_LF_PA_4000	mHS	LF	PA	4000	8
mHS_LF_PR_4000	mHS	LF	PR	4000	8
mHS_NS_FU_4000	mHS	NS	FU	4000	8
mHS_NS_PA_4000	mHS	NS	PA	4000	8
mHS_NS_PR_4000	mHS	NS	PR	4000	8

mHS accounts for two thirds of the corpus. Overall results weight each case equally, so mHS has twice the weight of HS. Family-specific results are shown separately. The source labels are balanced: 27,000 valid and 27,000 invalid cases.

Working inputs preserve the original question text, order, source identifiers, and all record fields, except the approved translations of `valido` to `valid` and `canonico` to `canonic` in the two LF/PR datasets. The analysis verifies every saved prompt and target against the frozen inputs and the Git source revision.

Which models answered?

The exact model IDs identify the experiment. Provider, service tier, and reasoning settings are part of each model condition. These differences prevent us from attributing differences to a model name or reasoning setting alone.

Model ID	Fixed endpoint	Tier	Reasoning requested	Effort
openai/gpt-4o-mini	openai	default	not supported	omitted
openai/gpt-6-luna	openai/flex	flex	disabled	omitted
anthropic/claude-haiku-5.5	anthropic	default	disabled	medium
anthropic/claude-sonnet-5.5	anthropic	default	enabled	low
google/gemini-3.1-flash-lite	google-ai-studio/flex	flex	disabled	omitted
google/gemini-3.5-flash-lite	google-ai-studio/flex	flex	enabled	minimal

- GPT-4o mini: archived model card
- GPT-6 Luna Flex: archived model card
- Claude Haiku 5.5: archived model card
- Claude Sonnet 5.5: archived model card
- Gemini 3.1 Flash-Lite Flex: archived model card
- Gemini 3.5 Flash-Lite Flex: archived model card

Providers were fixed and fallbacks disabled. The output limit was 1,024 tokens, including reasoning where the provider counts it in the output budget. Temperature, top-p, and seed were omitted. Responses are not deterministic. Caching and streaming were disabled.

The archived `earlier` and `later` tags are comparison slots. They are not equivalent generations across developers. Haiku and Sonnet are different model families. The main results therefore compare six individual models.

How are answers counted?

Measure	Numerator	Denominator
Source-label agreement	Matches	Interpretable answers
Interpretation coverage	Interpretable answers	All planned cases
Strict-format compliance	Only yes or no, ignoring case and outer whitespace	All planned cases
Matches over all cases	Matches	All planned cases

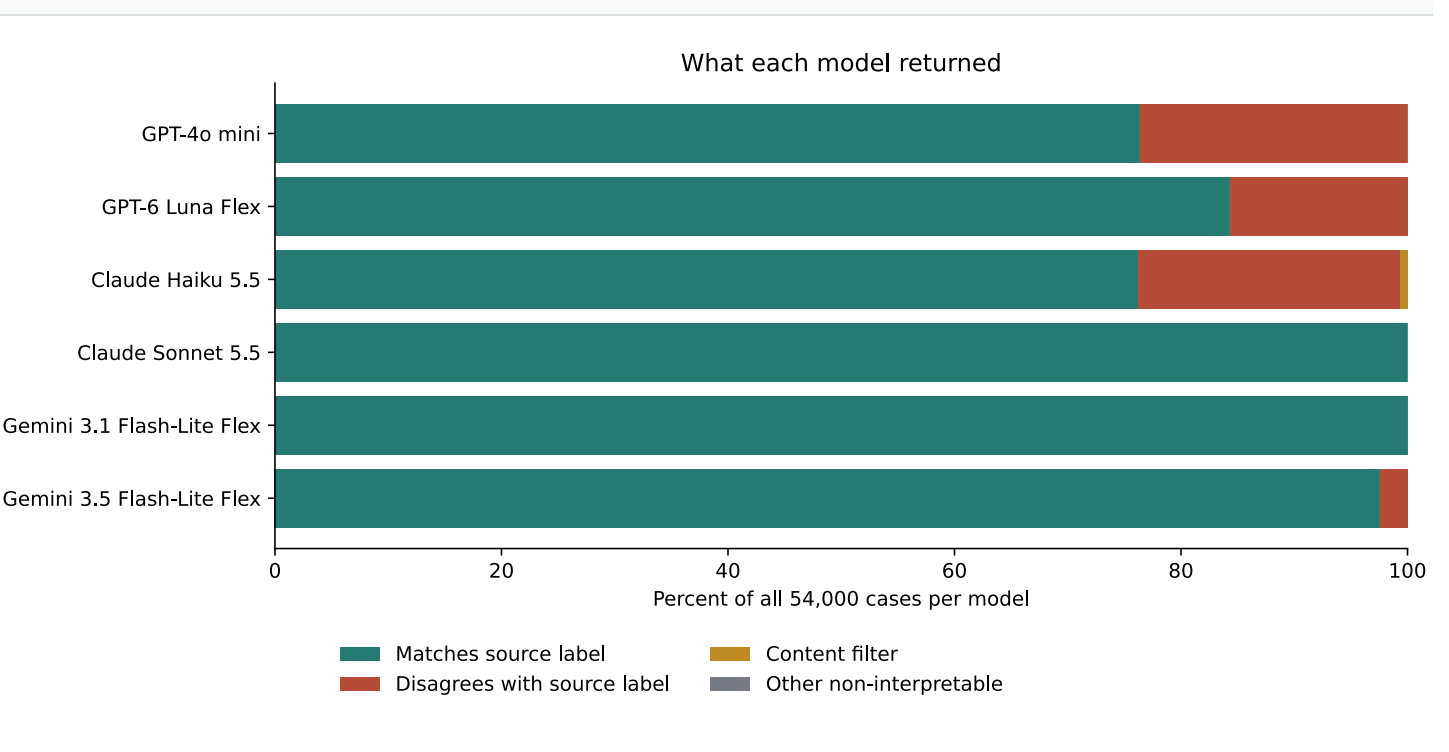
Scoring uses `source-label-deterministic-v3` and extraction rule `initial_or_final_line_v2`. It can read a supported yes/no answer at the beginning or on the final line. Explanations, punctuation, and bold answers may be interpretable but still fail the strict-format instruction. A response containing both complete terms yes and no is ambiguous. Errors, filters, empty responses, ambiguity, and unsupported formats keep their existing precedence. They are never converted to no.

A source-label disagreement is not automatically a logical error. A source-label match is not independent proof of logical validity. We do not change source labels or scores in this report.

What did each model return?

Model	Agreement / interpreted	Coverage / 54,000	Strict / 54,000	Matches / 54,000
GPT-4o mini	41,218/53,999 (76.33%)	53,999/54,000 (100.00%)	50,246/54,000 (93.05%)	41,218/54,000 (76.33%)
GPT-6 Luna Flex	45,490/54,000 (84.24%)	54,000/54,000 (100.00%)	54,000/54,000 (100.00%)	45,490/54,000 (84.24%)
Claude Haiku 5.5	41,186/53,648 (76.77%)	53,648/54,000 (99.35%)	53,642/54,000 (99.34%)	41,186/54,000 (76.27%)
Claude Sonnet 5.5	53,973/53,995 (99.96%)	53,995/54,000 (99.99%)	44,340/54,000 (82.11%)	53,973/54,000 (99.95%)
Gemini 3.1 Flash-Lite Flex	53,959/54,000 (99.92%)	54,000/54,000 (100.00%)	53,961/54,000 (99.93%)	53,959/54,000 (99.92%)
Gemini 3.5 Flash-Lite Flex	52,640/54,000 (97.48%)	54,000/54,000 (100.00%)	53,987/54,000 (99.98%)	52,640/54,000 (97.48%)

Model	Matches	Disagreements	Filters	Empty	Ambiguous	Unsupported format	Errors
GPT-4o mini	41,218	12,781	0	0	0	1	0
GPT-6 Luna Flex	45,490	8,510	0	0	0	0	0
Claude Haiku 5.5	41,186	12,462	352	0	0	0	0
Claude Sonnet 5.5	53,973	22	0	0	5	0	0
Gemini 3.1 Flash-Lite Flex	53,959	41	0	0	0	0	0
Gemini 3.5 Flash-Lite Flex	52,640	1,360	0	0	0	0	0



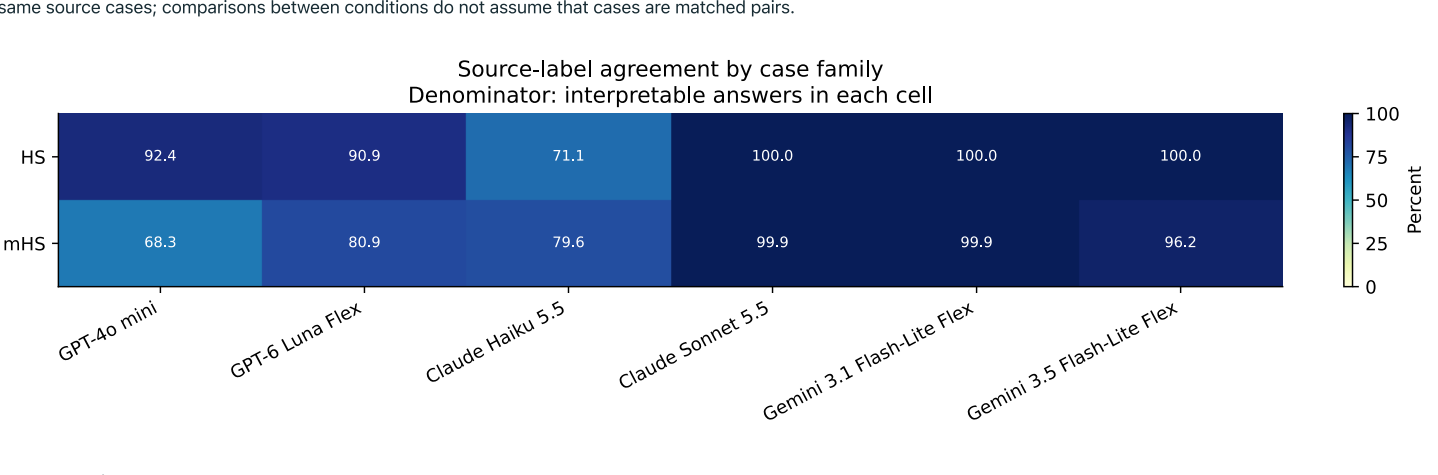
Yes and no answers

Model	Interpreted answer	Count
Claude Haiku 5.5	no	14718
Claude Haiku 5.5	yes	38930
Claude Sonnet 5.5	no	27021
Claude Sonnet 5.5	yes	26974
Gemini 3.1 Flash-Lite Flex	no	27041
Gemini 3.1 Flash-Lite Flex	yes	26959
Gemini 3.5 Flash-Lite Flex	no	26372
Gemini 3.5 Flash-Lite Flex	yes	27628
GPT-4o mini	no	37713
GPT-4o mini	yes	16286
GPT-6 Luna Flex	no	18526
GPT-6 Luna Flex	yes	35474

The source-label confusion table crosses the expected answer with the interpreted answer. It excludes non-interpretable responses, whose counts remain above.

Where do results differ?

The following tables compare observed results within the corpus. They do not establish causal effects. Each cell states its denominator. Comparisons between models use the same source cases; comparisons between conditions do not assume that cases are matched pairs.



Case family

The largest observed range across case family is for GPT-4o mini: 24.05 percentage points between HS (16,625/17,999 (92.37%)) and mHS (24,593/36,000 (68.31%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	HS	12,704/17,865 (71.11%)	17,865/18,000 (99.25%)	17,864/18,000 (99.24%)
Claude Haiku 5.5	mHS	28,482/35,783 (79.60%)	35,783/36,000 (99.40%)	35,778/36,000 (99.38%)
Claude Sonnet 5.5	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	12,526/18,000 (69.59%)
Claude Sonnet 5.5	mHS	35,973/35,995 (99.94%)	35,995/36,000 (99.99%)	31,814/36,000 (88.37%)
Gemini 3.1 Flash-Lite Flex	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.1 Flash-Lite Flex	mHS	35,959/36,000 (99.89%)	36,000/36,000 (100.00%)	35,964/36,000 (99.90%)
Gemini 3.5 Flash-Lite Flex	HS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
Gemini 3.5 Flash-Lite Flex	mHS	34,640/36,000 (96.22%)	36,000/36,000 (100.00%)	35,987/36,000 (99.96%)
GPT-4o mini	HS	16,625/17,999 (92.37%)	17,999/18,000 (99.99%)	16,483/18,000 (91.57%)
GPT-4o mini	mHS	24,593/36,000 (68.31%)	36,000/36,000 (100.00%)	33,763/36,000 (93.79%)
GPT-6 Luna Flex	HS	16,355/18,000 (90.86%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	mHS	29,135/36,000 (80.93%)	36,000/36,000 (100.00%)	36,000/36,000 (100.00%)

Lexical condition

The largest observed range across lexical condition is for GPT-6 Luna Flex: 16.49 percentage points between NS (16,234/18,000 (90.19%)) and LF (13,266/18,000 (73.70%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	HF	15,041/18,000 (83.56%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Claude Haiku 5.5	LF	12,611/17,924 (70.36%)	17,924/18,000 (99.58%)	17,920/18,000 (99.56%)
Claude Haiku 5.5	NS	13,534/17,724 (76.36%)	17,724/18,000 (98.47%)	17,723/18,000 (98.46%)
Claude Sonnet 5.5	HF	17,987/17,999 (99.93%)	17,999/18,000 (99.99%)	15,530/18,000 (86.28%)
Claude Sonnet 5.5	LF	17,997/18,000 (99.98%)	18,000/18,000 (100.00%)	14,165/18,000 (78.69%)
Claude Sonnet 5.5	NS	17,989/17,996 (99.96%)	17,996/18,000 (99.98%)	14,645/18,000 (81.36%)
Gemini 3.1 Flash-Lite Flex	HF	17,960/18,000 (99.78%)	18,000/18,000 (100.00%)	17,970/18,000 (99.83%)
Gemini 3.1 Flash-Lite Flex	LF	17,999/18,000 (99.99%)	18,000/18,000 (100.00%)	17,992/18,000 (99.96%)
Gemini 3.1 Flash-Lite Flex	NS	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	HF	17,469/18,000 (97.05%)	18,000/18,000 (100.00%)	17,989/18,000 (99.94%)
Gemini 3.5 Flash-Lite Flex	LF	17,590/18,000 (97.72%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	NS	17,581/18,000 (97.67%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
GPT-4o mini	HF	13,899/18,000 (77.22%)	18,000/18,000 (100.00%)	16,285/18,000 (90.47%)
GPT-4o mini	LF	13,357/18,000 (74.21%)	18,000/18,000 (100.00%)	16,725/18,000 (92.92%)
GPT-4o mini	NS	13,962/17,999 (77.57%)	17,999/18,000 (99.99%)	17,236/18,000 (95.76%)
GPT-6 Luna Flex	HF	15,990/18,000 (88.83%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	LF	13,266/18,000 (73.70%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	NS	16,234/18,000 (90.19%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

Verbal time

The largest observed range across verbal time is for Claude Haiku 5.5: 7.40 percentage points between PR (14,490/17,851 (81.17%)) and FU (13,139/17,810 (73.77%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	FU	13,139/17,810 (73.77%)	17,810/18,000 (98.94%)	17,805/18,000 (98.92%)
Claude Haiku 5.5	PA	13,557/17,987 (75.37%)	17,987/18,000 (99.93%)	17,986/18,000 (99.92%)
Claude Haiku 5.5	PR	14,490/17,851 (81.17%)	17,851/18,000 (99.17%)	17,851/18,000 (99.17%)
Claude Sonnet 5.5	FU	17,999/17,999 (100.00%)	17,999/18,000 (99.99%)	14,529/18,000 (80.72%)
Claude Sonnet 5.5	PA	17,986/17,999 (99.93%)	17,999/18,000 (99.99%)	14,545/18,000 (80.81%)
Claude Sonnet 5.5	PR	17,988/17,997 (99.95%)	17,997/18,000 (99.98%)	15,266/18,000 (84.81%)
Gemini 3.1 Flash-Lite Flex	FU	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,987/18,000 (99.93%)
Gemini 3.1 Flash-Lite Flex	PA	17,963/18,000 (99.79%)	18,000/18,000 (100.00%)	17,990/18,000 (99.94%)
Gemini 3.1 Flash-Lite Flex	PR	17,996/18,000 (99.98%)	18,000/18,000 (100.00%)	17,984/18,000 (99.91%)
Gemini 3.5 Flash-Lite Flex	FU	17,734/18,000 (98.52%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	PA	17,358/18,000 (96.43%)	18,000/18,000 (100.00%)	17,993/18,000 (99.96%)
Gemini 3.5 Flash-Lite Flex	PR	17,548/18,000 (97.49%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
GPT-4o mini	FU	14,096/18,000 (78.31%)	18,000/18,000 (100.00%)	16,660/18,000 (92.56%)
GPT-4o mini	PA	13,180/17,999 (73.23%)	17,999/18,000 (99.99%)	17,032/18,000 (94.62%)
GPT-4o mini	PR	13,942/18,000 (77.46%)	18,000/18,000 (100.00%)	16,554/18,000 (91.97%)
GPT-6 Luna Flex	FU	14,924/18,000 (82.91%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	PA	15,174/18,000 (84.30%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	PR	15,392/18,000 (85.51%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

Premise order

The largest observed range across premise order is for GPT-4o mini: 23.51 percentage points between canonic (23,783/27,000 (88.09%)) and non-canonic (17,435/26,999 (64.58%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	canonic	20,226/26,783 (75.52%)	26,783/27,000 (99.20%)	26,783/27,000 (99.20%)
Claude Haiku 5.5	non-canonic	20,960/26,865 (78.02%)	26,865/27,000 (99.50%)	26,859/27,000 (99.48%)
Claude Sonnet 5.5	canonic	26,973/26,995 (99.92%)	26,995/27,000 (99.98%)	23,585/27,000 (87.35%)
Claude Sonnet 5.5	non-canonic	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)	20,755/27,000 (76.87%)
Gemini 3.1 Flash-Lite Flex	canonic	26,967/27,000 (99.88%)	27,000/27,000 (100.00%)	26,968/27,000 (99.88%)
Gemini 3.1 Flash-Lite Flex	non-canonic	26,992/27,000 (99.97%)	27,000/27,000 (100.00%)	26,993/27,000 (99.97%)
Gemini 3.5 Flash-Lite Flex	canonic	26,143/27,000 (96.83%)	27,000/27,000 (100.00%)	26,987/27,000 (99.95%)
Gemini 3.5 Flash-Lite Flex	non-canonic	26,497/27,000 (98.14%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-4o mini	canonic	23,783/27,000 (88.09%)	27,000/27,000 (100.00%)	24,923/27,000 (92.31%)
GPT-4o mini	non-canonic	17,435/26,999 (64.58%)	26,999/27,000 (100.00%)	25,323/27,000 (93.79%)
GPT-6 Luna Flex	canonic	24,347/27,000 (90.17%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-6 Luna Flex	non-canonic	21,143/27,000 (78.31%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)

Source validity label

The largest observed range across source validity label is for Claude Haiku 5.5: 45.04 percentage points between valid (26,669/26,870 (99.25%)) and invalid (14,517/26,778 (54.21%)). This is a descriptive comparison.

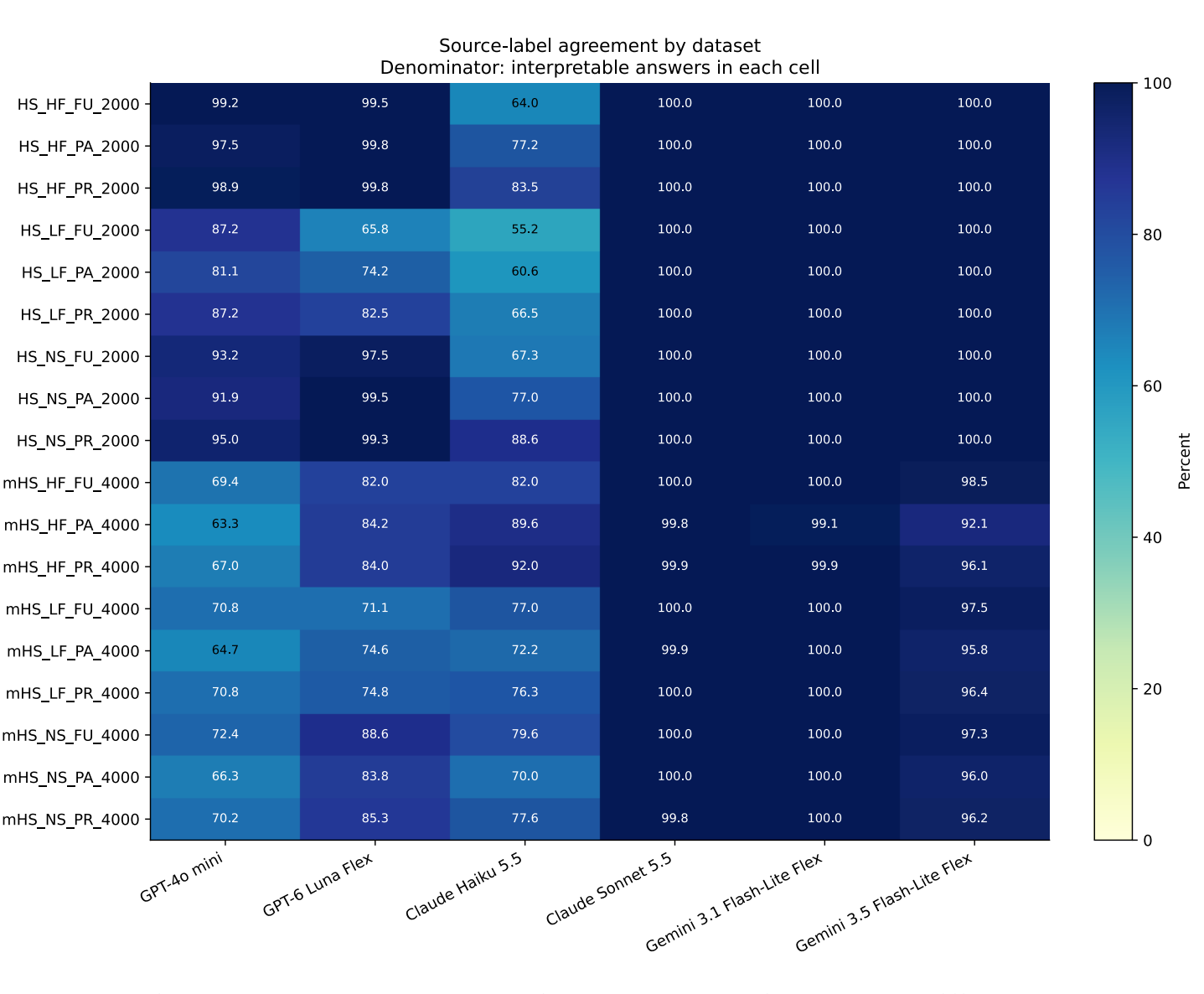
Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	invalid	14,517/26,778 (54.21%)	26,778/27,000 (99.18%)	26,773/27,000 (99.16%)
Claude Haiku 5.5	valid	26,669/26,870 (99.25%)	26,870/27,000 (99.52%)	26,869/27,000 (99.51%)
Claude Sonnet 5.5	invalid	26,999/26,999 (100.00%)	26,999/27,000 (100.00%)	17,600/27,000 (65.19%)
Claude Sonnet 5.5	valid	26,974/26,996 (99.92%)	26,996/27,000 (99.99%)	26,740/27,000 (99.04%)
Gemini 3.1 Flash-Lite Flex	invalid	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)	26,967/27,000 (99.88%)
Gemini 3.1 Flash-Lite Flex	valid	26,959/27,000 (99.85%)	27,000/27,000 (100.00%)	26,994/27,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	invalid	26,006/27,000 (96.32%)	27,000/27,000 (100.00%)	26,994/27,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	valid	26,634/27,000 (98.64%)	27,000/27,000 (100.00%)	26,993/27,000 (99.97%)
GPT-4o mini	invalid	25,966/27,000 (96.17%)	27,000/27,000 (100.00%)	26,132/27,000 (96.79%)
GPT-4o mini	valid	15,252/26,999 (56.49%)	26,999/27,000 (100.00%)	24,114/27,000 (89.31%)
GPT-6 Luna Flex	invalid	18,508/27,000 (68.55%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)
GPT-6 Luna Flex	valid	26,982/27,000 (99.93%)	27,000/27,000 (100.00%)	27,000/27,000 (100.00%)

Question formulation

The largest observed range across question formulation is for GPT-4o mini: 30.71 percentage points between if then / therefore (16,625/17,999 (92.37%)) and from the inference (therefore) / does the inference follow (11,099/18,000 (61.66%)). This is a descriptive comparison.

Model	Condition	Agreement	Coverage	Strict format
Claude Haiku 5.5	assuming that we can infer / thus	15,928/17,790 (89.53%)	17,790/18,000 (98.83%)	17,789/18,000 (98.83%)
Claude Haiku 5.5	from the inference (therefore) / does the inference follow	12,554/17,993 (69.77%)	17,993/18,000 (99.96%)	17,989/18,000 (99.94%)
Claude Haiku 5.5	if then / therefore	12,704/17,865 (71.11%)	17,865/18,000 (99.25%)	17,864/18,000 (99.24%)
Claude Sonnet 5.5	assuming that we can infer / thus	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	15,846/18,000 (88.03%)
Claude Sonnet 5.5	from the inference (therefore) / does the inference follow	17,973/17,995 (99.88%)	17,995/18,000 (99.97%)	15,968/18,000 (88.71%)
Claude Sonnet 5.5	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	12,526/18,000 (69.59%)
Gemini 3.1 Flash-Lite Flex	assuming that we can infer / thus	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,975/18,000 (99.86%)
Gemini 3.1 Flash-Lite Flex	from the inference (therefore) / does the inference follow	17,959/18,000 (99.77%)	18,000/18,000 (100.00%)	17,989/18,000 (99.94%)
Gemini 3.1 Flash-Lite Flex	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	17,997/18,000 (99.98%)
Gemini 3.5 Flash-Lite Flex	assuming that we can infer / thus	17,982/18,000 (99.90%)	18,000/18,000 (100.00%)	17,999/18,000 (99.99%)
Gemini 3.5 Flash-Lite Flex	from the inference (therefore) / does the inference follow	16,658/18,000 (92.54%)	18,000/18,000 (100.00%)	17,988/18,000 (99.93%)
Gemini 3.5 Flash-Lite Flex	if then / therefore	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-4o mini	assuming that we can infer / thus	13,494/18,000 (74.97%)	18,000/18,000 (100.00%)	16,673/18,000 (92.63%)
GPT-4o mini	from the inference (therefore) / does the inference follow	11,099/18,000 (61.66%)	18,000/18,000 (100.00%)	17,090/18,000 (94.94%)
GPT-4o mini	if then / therefore	16,625/17,999 (92.37%)	17,999/18,000 (99.99%)	16,483/18,000 (91.57%)
GPT-6 Luna Flex	assuming that we can infer / thus	15,956/18,000 (88.64%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	from the inference (therefore) / does the inference follow	13,179/18,000 (73.22%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)
GPT-6 Luna Flex	if then / therefore	16,355/18,000 (90.86%)	18,000/18,000 (100.00%)	18,000/18,000 (100.00%)

Datasets and complete strata



The dataset table contains all 108 model/dataset cells. The stratum results contain all 648 model/stratum cells. Join them to the stratum definitions by stratum. The HTML report supports combined condition filters.

Do models disagree on the same cases?

Each pair contains the same 54,000 UIDs. Agreement between answers and one-model-only source matches use only cases where both models gave interpretable answers. The excluded count remains visible. No missing answer is treated as a disagreement.

Model A	Model B	Same answer / both interpreted	Only A matches	Only B matches	Both match	Both disagree	Not both interpreted
GPT-4o mini	GPT-6 Luna Flex	33,610/53,999 (62.24%)	8059	12330	33159	451	1
GPT-4o mini	Claude Haiku 5.5	29,875/53,647 (55.69%)	11755	12017	29168	707	353
GPT-4o mini	Claude Sonnet 5.5	41,227/53,994 (76.35%)	4	12763	41209	18	6
GPT-4o mini	Gemini 3.1 Flash-Lite Flex	41,257/53,999 (76.40%)	1	12741	41217	40	1
GPT-4o mini	Gemini 3.5 Flash-Lite Flex	40,562/53,999 (75.12%)	1008	12429	40210	352	1
GPT-6 Luna Flex	Claude Haiku 5.5	41,911/53,648 (78.12%)	7857	3880	37306	4605	352
GPT-6 Luna Flex	Claude Sonnet 5.5	45,463/53,995 (84.20%)	22	8510	45463	0	5
GPT-6 Luna Flex	Gemini 3.1 Flash-Lite Flex	45,449/54,000 (84.16%)	41	8510	45449	0	0
GPT-6 Luna Flex	Gemini 3.5 Flash-Lite Flex	45,104/54,000 (83.53%)	873	8023	44617	487	0
Claude Haiku 5.5	Claude Sonnet 5.5	41,167/53,643 (76.74%)	18	12458	41163	4	357
Claude Haiku 5.5	Gemini 3.1 Flash-Lite Flex	41,167/53,648 (76.74%)	30	12451	41156	11	352
Claude Haiku 5.5	Gemini 3.5 Flash-Lite Flex	41,208/53,648 (76.81%)	669	11771	40517	691	352
Claude Sonnet 5.5	Gemini 3.1 Flash-Lite Flex	53,932/53,995 (99.88%)	41	22	53932	0	5
Claude Sonnet 5.5	Gemini 3.5 Flash-Lite Flex	52,625/53,995 (97.46%)	1354	16	52619	6	5
Gemini 3.1 Flash-Lite Flex	Gemini 3.5 Flash-Lite Flex	52,641/54,000 (97.48%)	1339	20	52620	21	0

Cases that make the results concrete

Examples are selected by the smallest SHA-256 hash of the UID for each family and category, without repeating a case. Family order is HS then mHS; category order is different answers, content filter, then interpretable non-strict format. These examples illustrate outcomes; they do not estimate prevalence. The HTML explorer exposes every case.

HS: different answers

UID: `HS_NS_FU_2000:955:ecd5d089ff37fde8`. Dataset: `HS_NS_FU_2000`; original row 955. Source label: `invalid`; expected answer: `no`.

> `If the preggar will glash the ploodnum, then the probeller will trelf the blorrack; if the probeller will trelf the blorrack, then the bruggant will frike the drambet; therefore if the bruggant will frike the drambet, then the preggar will glash the ploodnum?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	yes	disagreement	True
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	no	match	True

Claude Haiku 5.5 — original response

```
yes
```

Claude Sonnet 5.5 — original response

```
no
```

Gemini 3.1 Flash-Lite Flex — original response

```
no
```

Gemini 3.5 Flash-Lite Flex — original response

```
no
```

GPT-4o mini — original response

```
no
```

GPT-6 Luna Flex — original response

```
no
```

HS: content filter

UID: `HS_NS_PR_2000:37:7f42d42c0023eb28`. Dataset: `HS_NS_PR_2000`; original row 37. Source label: `valid`; expected answer: `yes`.

> `If the preggar gresks the prebart, then the brettent glashes the flendop; if the brettent glashes the flendop, then the probeller frikes the ploodnum; therefore if the preggar gresks the prebart, then the probeller frikes the ploodnum?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	not interpreted	content_filter	False
Claude Sonnet 5.5	yes	match	True
Gemini 3.1 Flash-Lite Flex	yes	match	True
Gemini 3.5 Flash-Lite Flex	yes	match	True
GPT-4o mini	yes	match	True
GPT-6 Luna Flex	yes	match	True

Claude Haiku 5.5 — original response

```
This request triggered restrictions on content related to the development of high-yield explosives and was blocked under Anthropic's Usage Policy. To learn more, provide feedback, or request an exemption based on how you use Claude, visit our help center: https://support.claude.com/ API integrators: you can reduce refusals for your users by configuring a fallback model — see https://platform.claude.com/docs/en/build-with-claude/refusals-and-fallback
```

Claude Sonnet 5.5 — original response

```
yes
```

Gemini 3.1 Flash-Lite Flex — original response

```
yes
```

Gemini 3.5 Flash-Lite Flex — original response

```
yes
```

GPT-4o mini — original response

```
yes
```

GPT-6 Luna Flex — original response

```
yes
```

HS: interpretable, non-strict format

UID: `HS_HF_FU_2000:1602:787e8fed73f8e1dc`. Dataset: `HS_HF_FU_2000`; original row 1602. Source label: `invalid`; expected answer: `no`.

> `If the student will hate the computer, then the family will see the college; if the doctor will win the game, then the student will hate the computer; therefore if the family will see the college, then the doctor will win the game?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	no	match	True
Claude Sonnet 5.5	no	match	False
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	no	match	True

Claude Haiku 5.5 — original response

```
no
```

Claude Sonnet 5.5 — original response

```
Let H = the student will hate the computer, F = the family will see the college, D = the doctor will win the game.

The premises are H → F and D → H. Chaining them gives D → F.

The conclusion asks for F → D, which is the converse of D → F. The premises do not support it. For example, if F is true while D and H are both false, both premises hold and the conclusion fails.

No
```

Gemini 3.1 Flash-Lite Flex — original response

```
no
```

Gemini 3.5 Flash-Lite Flex — original response

```
no
```

GPT-4o mini — original response

```
no
```

GPT-6 Luna Flex — original response

```
no
```

mHS: different answers

UID: `mHS_HF_PR_4000:1953:d7b258905cf06b38`. Dataset: `mHS_HF_PR_4000`; original row 1953. Source label: `invalid`; expected answer: `no`.

> `From the inference 'the woman pays the sugar, therefore the girl hates the school', and the inference 'the girl hates the school, therefore the officer finds the college', does the inference 'the officer finds the college, therefore the woman pays the sugar' follow?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	no	match	True
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	yes	disagreement	True

Claude Haiku 5.5 — original response

```
no
```

Claude Sonnet 5.5 — original response

```
no
```

Gemini 3.1 Flash-Lite Flex — original response

```
no
```

Gemini 3.5 Flash-Lite Flex — original response

```
no
```

GPT-4o mini — original response

```
no
```

GPT-6 Luna Flex — original response

```
yes
```

mHS: content filter

UID: `mHS_NS_FU_4000:1062:cbe3c3e2be728394`. Dataset: `mHS_NS_FU_4000`; original row 1062. Source label: `invalid`; expected answer: `no`.

> `Assuming that the bruggant will trelf the drambet, we can infer that the preggar will prant the ploodnum. Assuming that the preggar will prant the ploodnum, we can infer that the probeller will glash the prebart. Thus, assuming that the probeller will glash the prebart, can we infer that the bruggant will trelf the drambet?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	not interpreted	content_filter	False
Claude Sonnet 5.5	no	match	True
Gemini 3.1 Flash-Lite Flex	no	match	True
Gemini 3.5 Flash-Lite Flex	no	match	True
GPT-4o mini	no	match	True
GPT-6 Luna Flex	yes	disagreement	True

Claude Haiku 5.5 — original response

```
This request triggered restrictions on content related to the development of high-yield explosives and was blocked under Anthropic's Usage Policy. To learn more, provide feedback, or request an exemption based on how you use Claude, visit our help center: https://support.claude.com/ API integrators: you can reduce refusals for your users by configuring a fallback model — see https://platform.claude.com/docs/en/build-with-claude/refusals-and-fallback
```

Claude Sonnet 5.5 — original response

```
no
```

Gemini 3.1 Flash-Lite Flex — original response

```
no
```

Gemini 3.5 Flash-Lite Flex — original response

```
no
```

GPT-4o mini — original response

```
no
```

GPT-6 Luna Flex — original response

```
yes
```

mHS: interpretable, non-strict format

UID: `mHS_HF_PR_4000:153:99c4ec3b9f63ae06`. Dataset: `mHS_HF_PR_4000`; original row 153. Source label: `valid`; expected answer: `yes`.

> `Assuming that the family pays the computer, we can infer that the man loses the game. Assuming that the man loses the game, we can infer that the woman hates the school. Thus, assuming that the family pays the computer, can we infer that the woman hates the school?`

Model	Interpreted answer	Outcome	Strict format
Claude Haiku 5.5	yes	match	True
Claude Sonnet 5.5	yes	match	True
Gemini 3.1 Flash-Lite Flex	yes	match	True
Gemini 3.5 Flash-Lite Flex	yes	match	True
GPT-4o mini	yes	match	False
GPT-6 Luna Flex	yes	match	True

Claude Haiku 5.5 — original response

```
yes
```

Claude Sonnet 5.5 — original response

```
yes
```

Gemini 3.1 Flash-Lite Flex — original response

```
yes
```

Gemini 3.5 Flash-Lite Flex — original response

```
Yes
```

GPT-4o mini — original response

```
Yes.
```

GPT-6 Luna Flex — original response

```
yes
```

What did the run cost?

These are costs reported by OpenRouter, not an independent invoice audit. Costs are deduplicated by generation ID. A captured response cost takes precedence over a later lookup; conflicting observations remain in the database and conflict table. Zero is retained as zero. Missing costs remain unknown.

Phase	Model	Generation role	IDs with cost / all IDs	Unknown-cost IDs	Known USD
corpus	Claude Haiku 5.5	final	53648/54000	352	0.789836
corpus	Claude Sonnet 5.5	final	54000/54000	0	26.647214
corpus	Gemini 3.1 Flash-Lite Flex	final	54000/54000	0	0.615929
corpus	Gemini 3.5 Flash-Lite Flex	final	54000/54000	0	0.757996
corpus	GPT-4o mini	final	54000/54000	0	0.804368
corpus	GPT-6 Luna Flex	additional	0/426	426	unknown
corpus	GPT-6 Luna Flex	final	54000/54000	0	0.321379
preflight	Claude Haiku 5.5	final	1/1	0	0.000012
preflight	Claude Sonnet 5.5	final	1/1	0	0.000234
preflight	Gemini 3.1 Flash-Lite Flex	final	1/1	0	0.000009
preflight	Gemini 3.5 Flash-Lite Flex	final	1/1	0	0.000011
preflight	GPT-4o mini	final	1/1	0	0.000012
preflight	GPT-6 Luna Flex	final	1/1	0	0.000005

Across this run's corpus and six-call preflight, known reported cost is **USD 29.937005**. There are **778 generation IDs with unknown cost** and **0 conflicting cost observations**. Amounts from earlier runs are excluded.

Final generations have an explicit final response ID. Additional generations are other identified attempts for a sample with a final ID. Unresolved generations belong to samples without an explicit final ID; their role is not guessed. Attempts without IDs cannot be assigned a deduplicated cost.

Phase	Model	HTTP attempts	Failed attempts	Attempts without ID	Without end timestamp
corpus	Claude Haiku 5.5	54000	0	0	0
corpus	Claude Sonnet 5.5	54000	0	0	0
corpus	Gemini 3.1 Flash-Lite Flex	54001	1	1	0
corpus	Gemini 3.5 Flash-Lite Flex	54000	0	0	0
corpus	GPT-4o mini	54000	0	0	0
corpus	GPT-6 Luna Flex	54426	426	0	0
preflight	Claude Haiku 5.5	1	0	0	0
preflight	Claude Sonnet 5.5	1	0	0	0
preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	0
preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	0
preflight	GPT-4o mini	1	0	0	0
preflight	GPT-6 Luna Flex	1	0	0	0

A failed HTTP attempt has no response, HTTP status 4xx/5xx, or an error payload inside HTTP 200. The count of attempts beyond one per sample measures additional HTTP attempts; it does not add new cases. A filter is a recorded response outcome and does not by itself trigger a replacement generation.

Tokens in final responses

Phase	Model	Input tokens	Output tokens	Reasoning tokens	Unknown input/output/reasoning samples
corpus	Claude Haiku 5.5	6894599	200752	0	352/352/352
corpus	Claude Sonnet 5.5	6996312	1265459	0	0/0/0
corpus	Gemini 3.1 Flash-Lite Flex	4603200	54039	0	0/0/0
corpus	Gemini 3.5 Flash-Lite Flex	4603200	54013	0	0/0/0
corpus	GPT-4o mini	5131440	57754	0	0/0/0
corpus	GPT-6 Luna Flex	5077440	270030	0	0/0/0
preflight	Claude Haiku 5.5	96	4	0	0/0/0
preflight	Claude Sonnet 5.5	97	4	0	0/0/0
preflight	Gemini 3.1 Flash-Lite Flex	65	1	0	0/0/0
preflight	Gemini 3.5 Flash-Lite Flex	65	1	0	0/0/0
preflight	GPT-4o mini	74	1	0	0/0/0
preflight	GPT-6 Luna Flex	73	5	0	0/0/0

Token totals come from final output usage. They exclude usage missing from that field and do not count the usage of failed attempts as zero. Reported zero reasoning tokens do not prove the absence of unreported internal processing.

What conclusions are supported?

The results describe these six model/provider/reasoning conditions on this fixed corpus in one execution. Interpret agreement together with coverage and strict-format compliance. Review the condition tables before using the overall ranking: mHS has twice the weight of HS.

Models can differ on the same question, and the paired table quantifies those differences without changing the cases. Filters and unparsed responses describe limits in usable coverage. Explanatory text that is interpreted successfully can still violate the requested format.

Next, review source labels for cases with widespread disagreement and inspect mHS and reversed-order formulations. A logical audit should be a separate versioned analysis. Repeated independent runs would be needed to measure response variability. This report includes no confidence intervals, significance tests, causal claims, or cross-pilot comparisons.

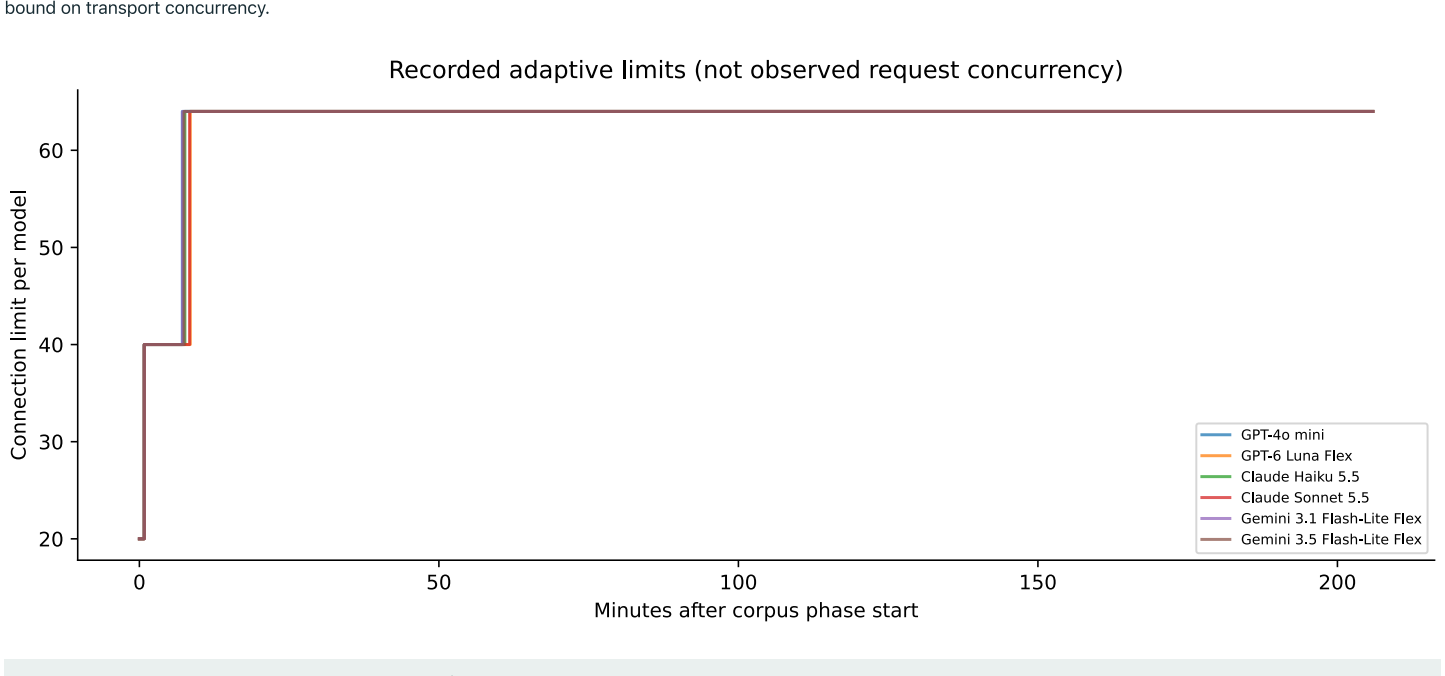
Operational appendix: time and concurrency

Interval	Minutes
Preparation and dispatch after corpus start	0.72
First task start to last task completion	205.20
Finalization and exports after last task	15.61
Full corpus phase including closure	221.52

Corpus phase started at 2026-11-06 01:24:17 -03 and closed at 2026-11-06 05:05:49 -03. Task duration includes generation, waits, retries, metadata collection, and scoring. It is not pure provider processing time. The preflight is separate.

The run allowed six active tasks and 32 samples per task, for a 192-slot configured ceiling. Adaptive limits were min 10, start 20, max 64 per model. The controller can reduce limits after rate limits; transient errors pause growth. Limits are not measured request concurrency.

The observed peak was **192 concurrent samples** and **140 concurrent captured HTTP intervals with both timestamps**. Missing HTTP end timestamps prevent a complete bound on transport concurrency.



Model	Sample peak	HTTP interval peak	Mean sample seconds	P90 sample seconds	Mean HTTP seconds	P90 HTTP seconds
GPT-4o mini	32	32	6.56	10.00	1.19	3.86
GPT-6 Luna Flex	64	56	7.68	11.73	2.25	6.04
Claude Haiku 5.5	32	32	7.08	10.57	1.69	4.54
Claude Sonnet 5.5	64	53	7.76	11.34	2.46	5.35
Gemini 3.1 Flash-Lite Flex	32	32	6.48	9.86	1.12	3.71
Gemini 3.5 Flash-Lite Flex	32	32	6.79	10.26	1.41	4.13

Request and returned metadata checks

Phase	Model	Samples	Missing captured requests	Samples with condition faults	Samples with reasoning usage reported	Nonzero reasoning reported
corpus	Claude Haiku 5.5	54000	0	0	53648	0
corpus	Claude Sonnet 5.5	54000	0	0	54000	0
corpus	Gemini 3.1 Flash-Lite Flex	54000	0	0	54000	0
corpus	Gemini 3.5 Flash-Lite Flex	54000	0	0	54000	0
corpus	GPT-4o mini	54000	0	0	54000	0
corpus	GPT-6 Luna Flex	54000	0	0	54000	0
preflight	Claude Haiku 5.5	1	0	0	1	0
preflight	Claude Sonnet 5.5	1	0	0	1	0
preflight	Gemini 3.1 Flash-Lite Flex	1	0	0	1	0
preflight	Gemini 3.5 Flash-Lite Flex	1	0	0	1	0
preflight	GPT-4o mini	1	0	0	1	0
preflight	GPT-6 Luna Flex	1	0	0	1	0

The returned routing table preserves reported provider and tier values, including empty sets. Missing evidence is not proof of a setting. The SQLite conditions table retains request-level fault descriptions. Native logs retain complete payloads and fields that this report does not project.

Evidence, definitions, and reproduction

Summary JSON · Validation record · Source and analysis hashes · SQLite analysis · Response table · Generation ledger · Cost conflicts

The report uses only the closed full-corpus run and its separate preflight. Inputs, original responses, native scores, logs, and exports are preserved. Every sample links to its native log and hash in the response table; every case includes source row, original identifiers, and record hashes in the dataset JSON files.

To rebuild offline, run `uv run --locked transitivity report --run results/full/2026-11-06_01-23-40-040fd7b7 --output <new-directory>`. The destination must not exist. No provider API requests are made. To browse the HTML bundle, serve the repository with `uv run --locked python -m http.server 8000` and open this folder's `index.html` through localhost. All runtime assets and case shards are local.